

Personalized Movie Recommendation System

Learning user taste from the Netflix Prize dataset to predict ratings and generate explainable Top-10 recommendations.

100.5M

ratings · 480K users

1.18%

matrix density

0.8505

best RMSE (SVD)

0.0610

best MAP@10 (Hybrid)

01 Problem Overview

Discovery at scale.

A streaming catalogue holds ~17,770 titles — no user can browse them all. The platform has to learn each person's taste and surface the few films they're most likely to enjoy.

THE CORE CHALLENGE

Sparsity

The user × movie matrix is **~99% empty**: 8.53 billion possible cells, only ~100M actually filled. We can't store it densely or train on all of it — so we use a faithful, sparsity-preserving subset.

The system must:



Learn preferences

from each user's rating history



Predict unseen ratings

measured by RMSE



Generate a Top-10 list

measured by MAP@10 (relevant ≥ 3.5★)



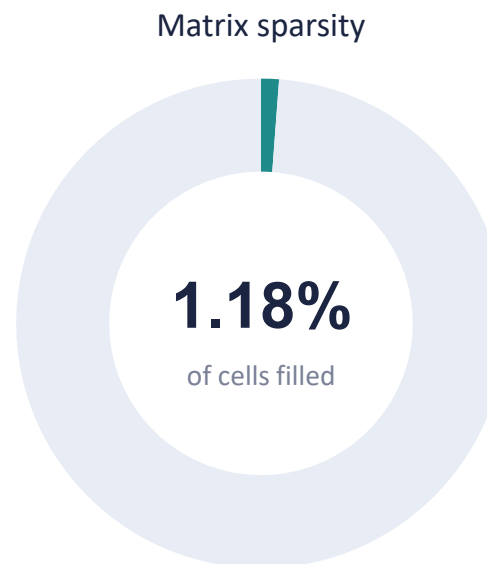
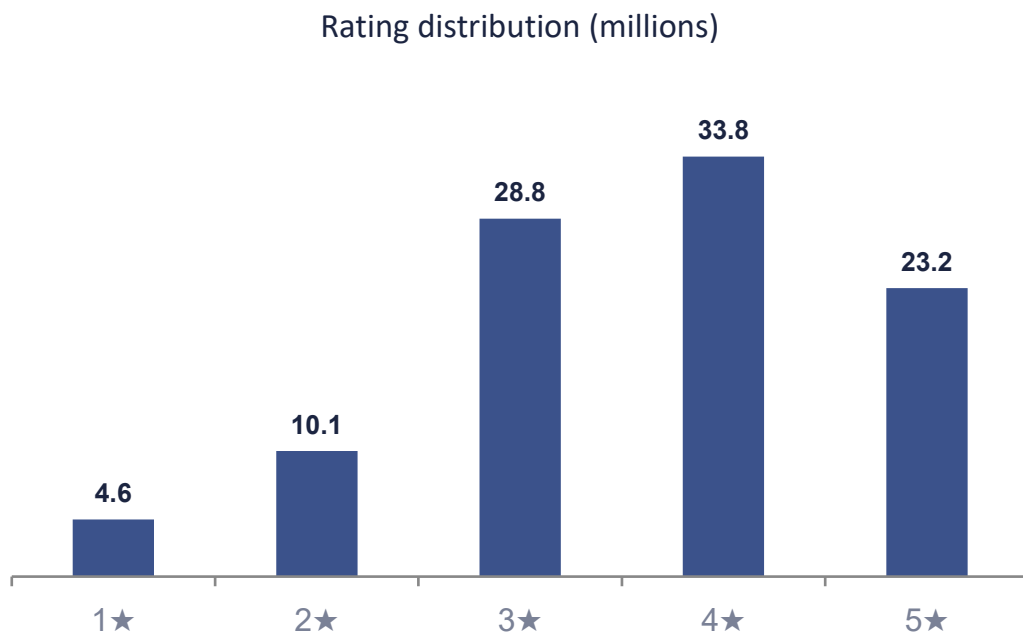
Explain its picks

honest “because you liked...” reasons

02

What the Data Tells Us

Exploratory analysis of ~100M ratings — and what each finding means for the model



Positive skew

Mean 3.604 · 56.6% are 4– or 5★

→ *bias terms matter; beat the average*

Long tail

Median user 96 · median movie 561

→ *item–item CF is the reliable axis*

Upward drift

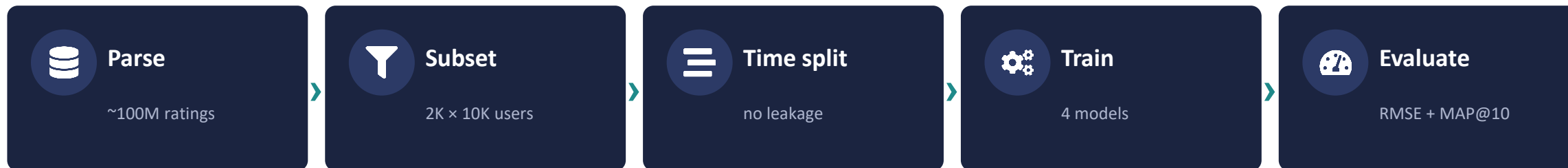
Mean rating rises over the years

→ *split by time, never randomly*

Takeaway: positive skew + long tail + 99% sparsity point straight to item-based collaborative filtering and matrix factorization (SVD), evaluated on a time-based split.

03 Approach

Simple, honest, explainable — a clear pipeline over clever-but-opaque tricks



1

Baseline

Bias only

A fair yardstick: μ + user + movie bias. Everything must beat it.

2

Item-KNN

Neighbourhood CF

"People who liked these also liked..." — naturally explainable.

3

SVD

Matrix factorization

Learns hidden taste features; copes with sparsity. The main ranker.

4

Hybrid

Content re-ranker

Title TF-IDF re-orders SVD; can rank brand-new movies too.

Evaluation: per-user time split (hold out each user's most recent 20%) · RMSE for accuracy · MAP@10 for ranking (relevant when true rating ≥ 3.5).

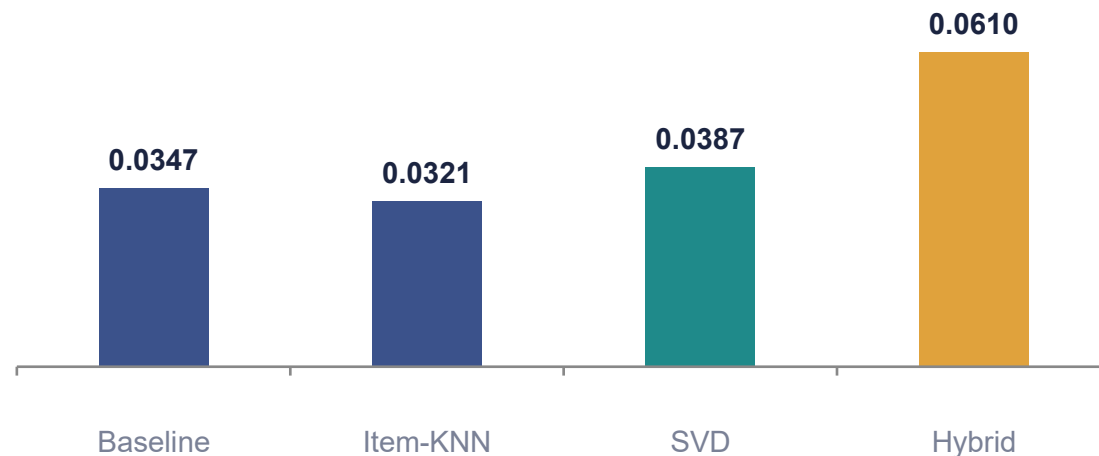
04

Experimental Results

SVD wins on rating accuracy; a lightweight hybrid wins on ranking quality

Model	RMSE ↓	MAP@10 ↑
Baseline	0.9078	0.0347
Item-based KNN	0.9653	0.0321
SVD	0.8505	0.0387
Hybrid (re-ranked SVD)	0.8505	0.0610

MAP@10 — ranking quality (higher is better)



+57.6%

ranking gain — Hybrid vs SVD MAP@10

26.1%

of the catalogue reached (healthy diversity)

0.8505

best RMSE — SVD beats the baseline on both

05

Recommendations in Action

The best model's Top-10, inspected against what each user already liked



SUCCESS

User 1621025 · AP@10 = 0.890

Liked: West Side Story · The Untouchables · Raiders of the Lost Ark · LOTR & Star Wars titles

Recommended: Empire Strikes Back ✓ · LOTR: Fellowship ✓ · A New Hope ✓ · Return of the King ✓ · The Simpsons S4 ✓ ...

9 of 10 hit. SVD locks onto the user's sci-fi/fantasy cluster and reaches titles plain neighbourhood CF couldn't.



INSTRUCTIVE FAILURE

User 1045204 · AP@10 = 0.000

Liked: rom-coms — How to Lose a Guy in 10 Days · Two Weeks Notice · Sister Act

Recommended: LOTR · CSI seasons · Shrek — 0 hits

Why (and why it's not as bad as it looks): taste drift to a new interest · popularity pull on thin history · MAP@10 only credits movies the user happened to rate.

Explainability: “Because you rated LOTR and Raiders 5 ★, you'll likely enjoy The Empire Strikes Back” — reasons come from real KNN co-rating similarity.



Bias terms do most of the work

Knowing who rates and what is rated explains most variance — judge models by lift over the baseline.



Sparsity drove model choice

At ~99% empty, SVD wins by generalizing across the blanks instead of needing shared raters.



RMSE $\neq$ ranking quality

The hybrid matches SVD's RMSE yet ranks far better — so we prioritized MAP@10, the discovery metric.



Time matters

Upward drift makes a time split essential; “changing tastes” is the main reason recommendations miss.



Explanations were nearly free

Reusing the KNN similarity matrix gave trustworthy “because you liked...” reasons with no extra training.

BEYOND THE BASICS



Explainable layer

SVD ranks, KNN explains



Cold-start strategy

damped popularity + popularity $\leftrightarrow$ SVD blend



Hybrid re-ranker

TF-IDF titles re-order SVD

Conclusion & What's Next

A clear, correct, explainable recommender — and a roadmap to push it further.

0.8505

best RMSE (SVD)

0.0610

best MAP@10 (Hybrid)

+57.6%

ranking gain from hybrid

WHAT I'D DO NEXT

- **SVD++ / time-aware SVD**

use implicit feedback and model rating drift directly

- **Neural Collaborative Filtering**

non-linear patterns, with SVD as a fast candidate generator

- **Ranking-first training (BPR)**

optimize the Top-10 directly instead of via rating prediction



Reproducible by design

One config, fixed seed = 42 for deterministic results.

GitHub repo + 7-page technical report.

Standalone pipeline: 01_prepare → 02_train →
03_evaluate → 04_recommend